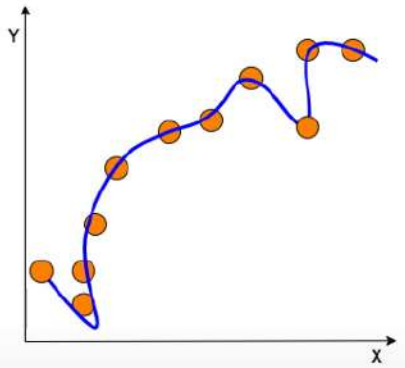
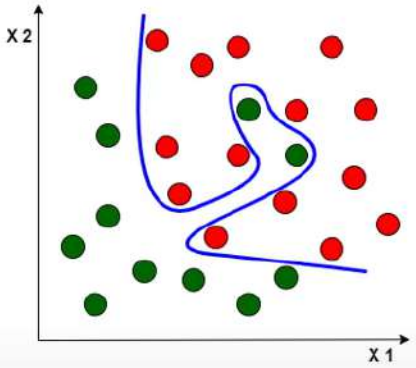
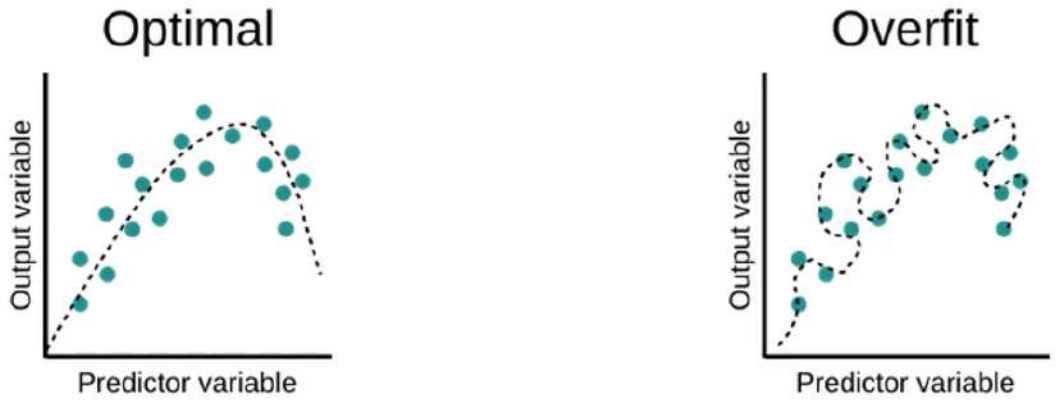
Overfitting in Machine Learning





Overfitting

Overfitting refers to a model that models the training data too well. Overfitting happens when a model learns the detail and noise in the training dataset to the extent that it negatively impacts the performance of the model.



Sign that the model has Overfitted : High Training data Accuracy & very low Test data Accuracy

Overfitting

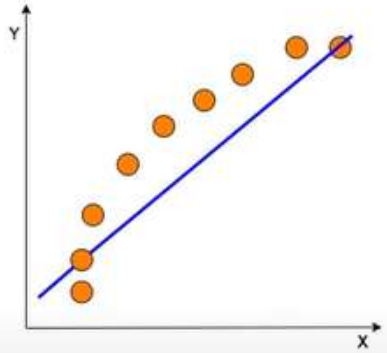
Causes for Overfitting:

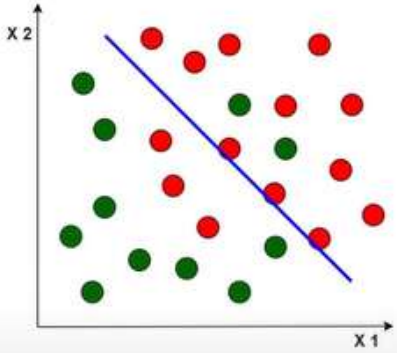
1. Less Data
2. Increased Complexity of the model
3. More number of layers in Neural Network

Preventing Overfitting by:

1. Using more data
2. Reduce the number of layers in the Neural network
3. Early Stopping
4. Bias – Variance Tradeoff
5. Use Dropouts

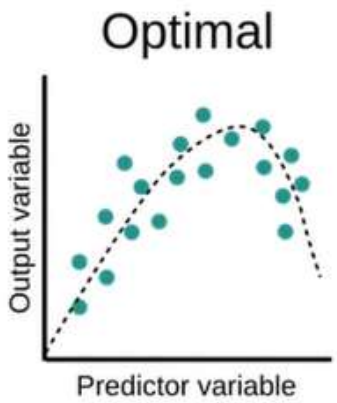
Underfitting in Machine Learning

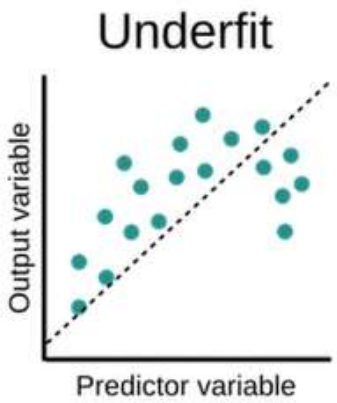




Underfitting

Underfitting happens when the model **does not learn enough** from the data. Underfitting occurs when a machine learning model cannot capture the underlying trend of the data





Sign that the model has Underfitted : Very Low Training data Accuracy

Underfitting

Causes for Underfitting:

1. Choosing a wrong model
2. Less complexity of the model
3. Less variance but high bias

Prevent Underfitting by:

1. Choosing the correct model appropriate for the problem
2. Increasing the complexity of the model
3. More number of parameters to the model
4. Bias – Variance Tradeoff